

CORTICOSTEROIDS

_____Name

_____Date

also called... **glucocorticosteroids, glucocorticoids, or cortisones.**

What are they?

Corticosteroids are anti-inflammatory compounds used to treat the skin, eye, ear, and nose. Sometimes they are used on the rectum or genital area. They may also be taken by injection, inhaler, or pills for conditions like asthma, arthritis, and colitis. Low strength hydrocortisone is available over-the-counter. Stronger cortisones require a prescription.

Corticosteroids are arranged into **five classes by structure**. If you are allergic to a drug in one group, you will likely react to the others in that class. Some classes also can show cross-reactivity with each other. (See table on the next page for a summary.)

A - Persons with a positive patch test to tixocortol-21-pivalate are allergic to steroids in Class A. This group includes some forms of hydrocortisone and prednisone. Class A sensitive individuals also often react to class D₂ steroids, which include some other hydrocortisones and prednicarbate. Safe substitutes are drugs in Classes B, C, and D₁.

B - Individuals who have positive patch tests to budesonide or triamcinolone acetonide should avoid corticosteroids in both Class B and D₂, listed on the next page. Safe alternatives should be chosen from Classes A, C, and D₁.

C - This class is rarely associated with allergy. It includes topical desoximetasone and clocortolone pivalate, as well as dexamethasone. Patients allergic to this group can usually safely use topical steroids in all the other classes.

D₁ - Class D corticosteroids are subdivided into two groups. Persons allergic to clobetasol-17-propionate must avoid all agents in the D₁ group. This includes alclometasone dipropionate, betamethasone dipropionate, betamethasone valerate, clobetasol propionate, diflorasone diacetate, fluticasone propionate, halobetasol propionate, and mometasone furoate. Reactors to this class should probably avoid D₂ steroids as well. Look to Class A, B, or C for less risky substitutes.

D₂ - Persons with a positive patch test to hydrocortisone-17-butyrate are allergic to Class D₂ cortisones, which are prednicarbate, hydrocortisone valerate, hydrocortisone probutate, and hydrocortisone butyrate. Class D₂ reactors should also avoid members of Classes D₁, A, and B. In other words, the safest steroids to use are those in Class C.

It is very important to tell your doctors and nurses about your allergy. Read the complete ingredient list of each product you use on your skin to look for the names on your allergy list. Look for the ingredients of all prescription medicines used on your skin, eyes, ears, nose, and genital area. Any cortisone injection or pill should also be checked against your allergy list.

NOTE: Topical steroids can remain in clothing even after laundering. Consider discarding items you have worn against treated sites, like knit socks, gloves, bras, two-ply garments, or shoes, since such garments may keep exposing your skin to your allergen!

CORTICOSTEROID CLASSES

<u>Class:</u>	A	B	C	D₁	D₂
Structure:	No D ring subs'n, except C21 short chain esters	C16, 17-cis-diol or –ketal	C16 -methyl substitution	C16 -methyl substitution, C17/21-long chain ester	C16-no methyl & no halogen, C17-long chain ester, +/- C21 chain
Testing Substance:	Tixocortol-21-pivalate	Budesonide/ Triamcinolone acetonide	Desoximetasone	Clobetasol-17-propionate	Hydrocortisone-17-butyrate
Usual Cross Reactions:	D ₂	D ₂	--	--	A & Budesonide
Members:	Cloprednol Dichlorisone acetate Fludrocortisone acetate Fluorometholone Fluprednisolone acetate Hydrocortisone Hydrocortisone acetate Medrysone Meprednisone Methylprednisolone Prednisolone Prednisone	Amcinonide Budesonide Desonide Flunisolide Fluocinolone acetonide Fluocinonide Flurandrenolide Halcinonide Triamcinolone Triamcinolone acetonide Triamcinolone diacetate	Betamethasone Betamethasone-21-disodium phosphate Clot cortolone pivalate Desoximetasone Dexamethasone Diflucortolone pivalate Diflucortolone valerate Flumethasone Fluocortin butyl ester Fluocortolone (hexanoate, pivalate, caproate) Fluprednidene acetate Halometasone	Alclometasone dipropionate Betamethasone dipropionate Betamethasone valerate Clobetasone butyrate Clobetasol propionate Diflorasone diacetate Fluticasone propionate Halobetasol propionate Mometasone furoate	Hydrocortisone buteprate Hydrocortisone butyrate Hydrocortisone probutate Hydrocortisone valerate Methylprednisolone aceponate Prednicarbate
Safe Substitutes:	B, C, D ₁	A, C, D ₁	A, B, D ₁ , D ₂	A, B, C	C